

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		$10.5 - 4.5 = 6.0$ [cm]		1		1	1	1
		(ii)		Weight of toy = 3.0 [N] (1) from graph / leaflet Division by 10 (1) $m = \frac{3.0}{10} = 0.3$ [kg] (1)	1	1 1		3	3	3
	(b)	(i)		If object A exerts a force on object B, then object B exerts an equal (1) and opposite force on object A (1) Alternative: Action and reaction force are equal (1) and opposite (1)	2			2		2
		(ii)		4 [N] (1) Up (1)		2		2		2
	(c)	(i)		Substitution: $W = 0.5 \times 35 \times 0.32$ (1) $= 5.6$ [J] (1)	1	1		2	2	
		(ii)		[Force doubles or force is 70 N and] the extension doubles or extension is 0.64 m or the work done is 11.2 [J] ecf (1) $W = 0.5 \times 70 \times 0.64 = 22.4$ [J] or implied (1) which is 4 times the WD or which isn't double or which isn't 11.2 [J] ecf (1) so the gym instructor is not correct. Conclusion must be present to award 3 marks			3	3	3	
				Question 8 total	4	6	3	13	9	8